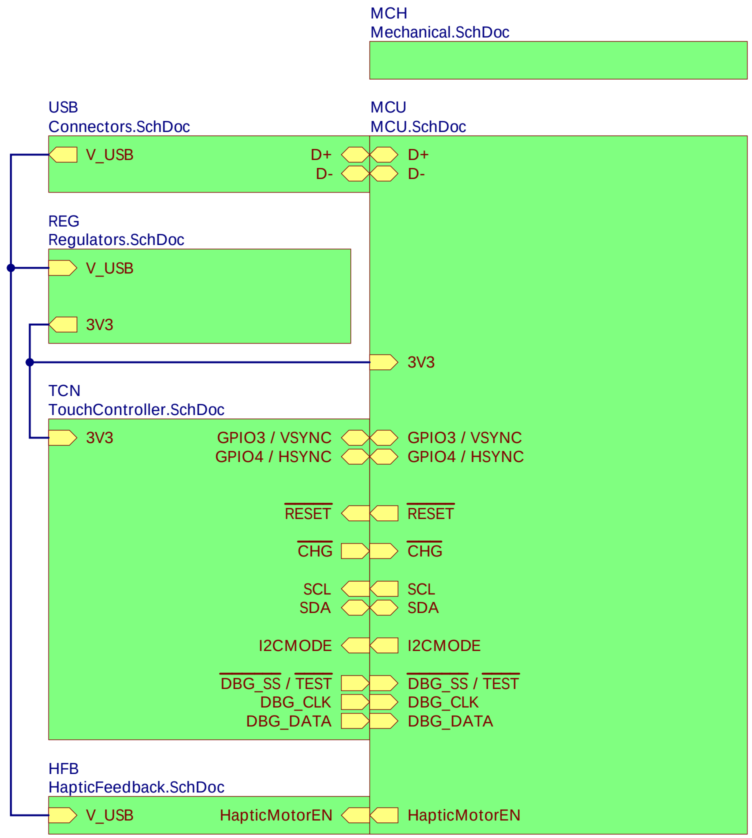

Title: Trackpad.SchDoc

PCB: Trackpad.PrjPCB

Version: R2.002

Date: 10/23/2024 Time: 9:31:17 PM

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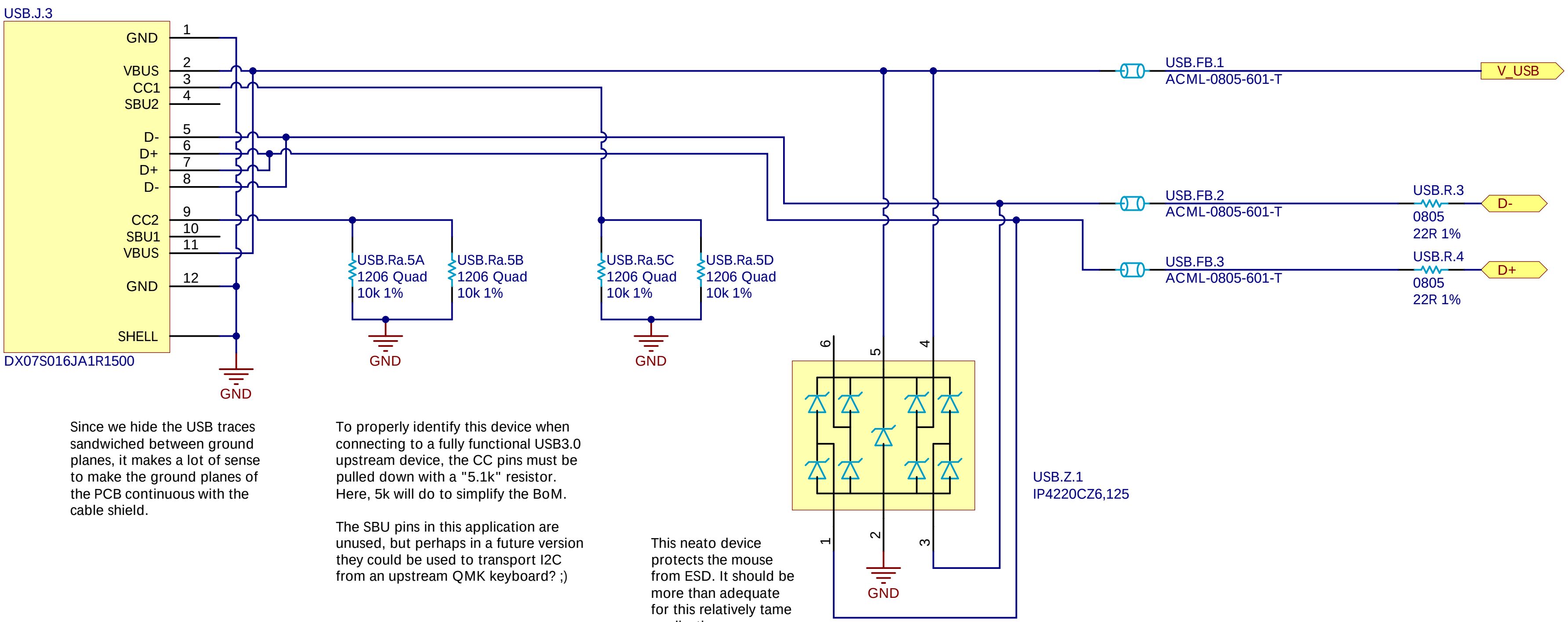

Since we hide the USB traces sandwiched between ground planes, it makes a lot of sense to make the ground planes of the PCB continuous with the cable shield.

To properly identify this device when connecting to a fully functional USB3.0 upstream device, the CC pins must be pulled down with a "5.1k" resistor. Here, 5k will do to simplify the BoM.

The SBU pins in this application are unused, but perhaps in a future version they could be used to transport I2C from an upstream QMK keyboard? ;)

This neato device protects the mouse from ESD. It should be more than adequate for this relatively tame application.

Title: Connectors.SchDoc
PCB: Trackpad.PrjPCB
Version: R2.002
Date: 10/23/2024 Time: 9:31:17 PM
Sheet 3 of 7

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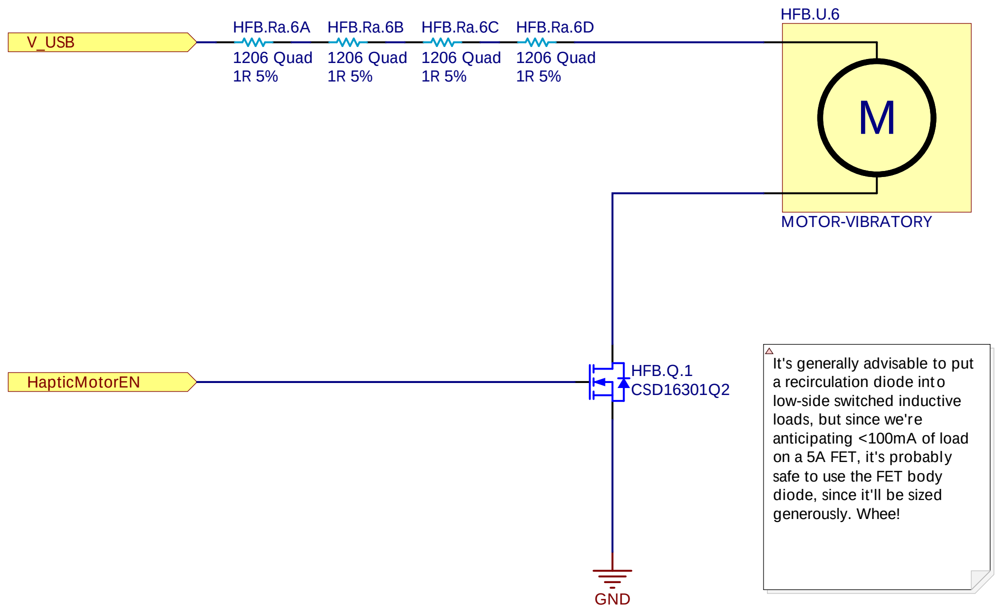

To keep things from getting too gnarly in the event of unexpected behaviour, we'll put an input resistor of 4R here. This provides cheap damping and current limiting.

It's generally advisable to put a recirculation diode into low-side switched inductive loads, but since we're anticipating <100mA of load on a 5A FET, it's probably safe to use the FET body diode, since it'll be sized generously. Whee!

Title: HapticFeedback.SchDoc			
PCB: Trackpad.PrjPCB		Version: R2.002	
Date: 10/23/2024	Time: 9:31:17 PM	Sheet 4	of 7

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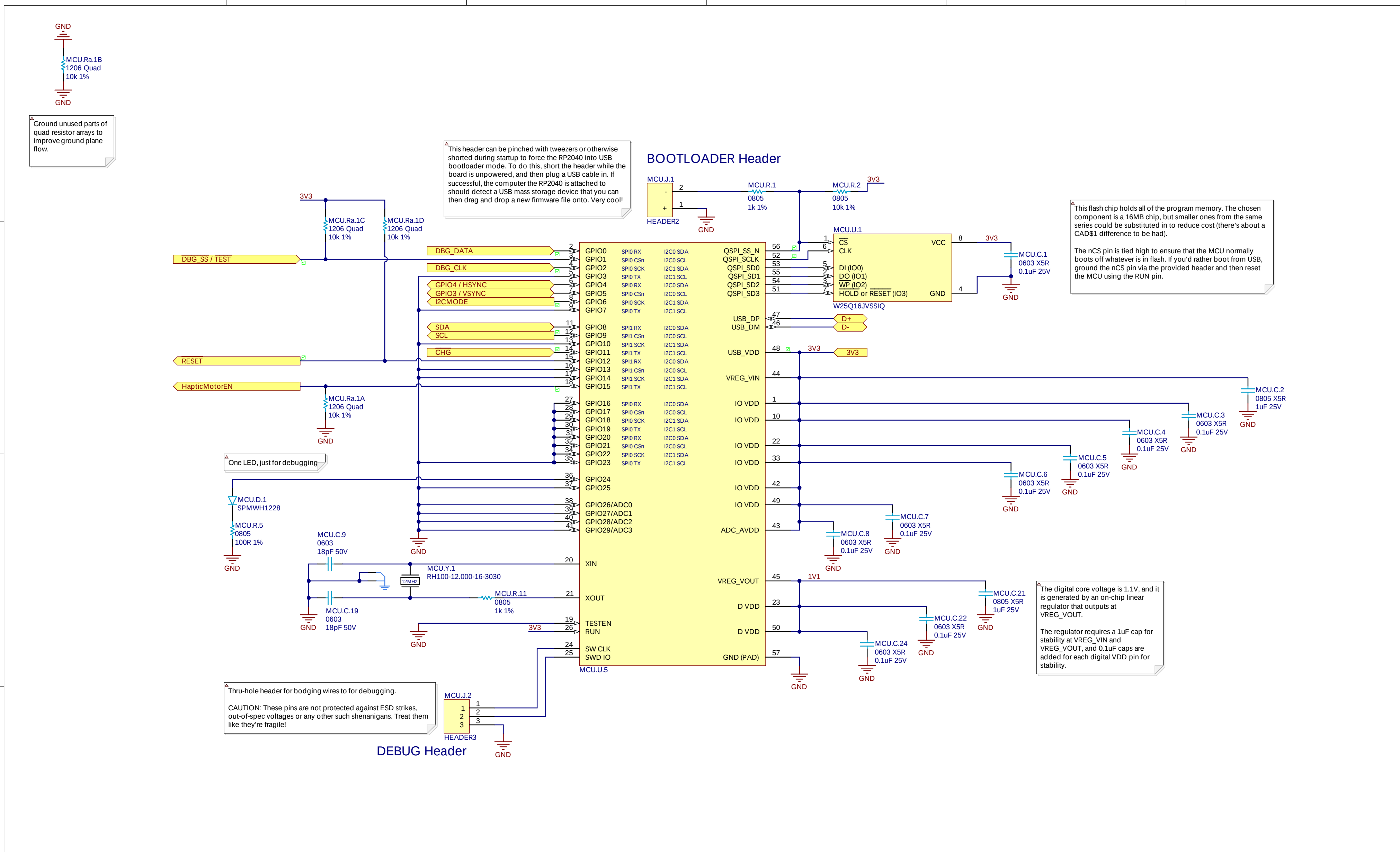

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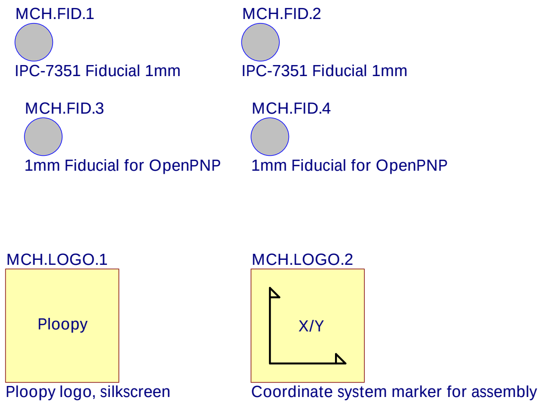
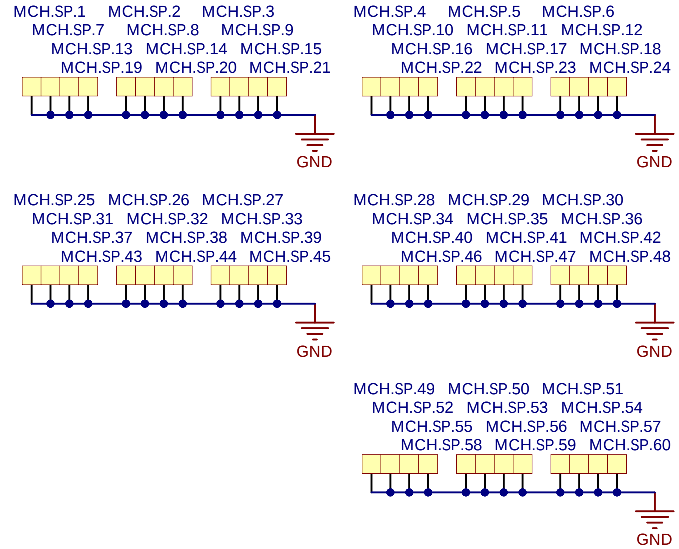
Title: MCU.SchDoc			Ploopy Corporation 137-1140 Burnhamthorpe Road W Mississauga ON L5C 0A3 Canada
PCB: Trackpad.PrjPCB		Version: R2.002	
Date: 10/23/2024	Time: 9:31:17 PM	Sheet 5 of 7	

Pick and Place Fiducials



Spark Gaps -- Case

Since the case has gaps in it, we expect ESD to worm its way in via creepage and perhaps other ways. To protect the board from this eventuality, we place spark gaps along the edges.



Title: Mechanical.SchDoc			
PCB: Trackpad.PrjPCB		Version: R2.002	
Date: 10/23/2024	Time: 9:31:17 PM	Sheet 6	of 7

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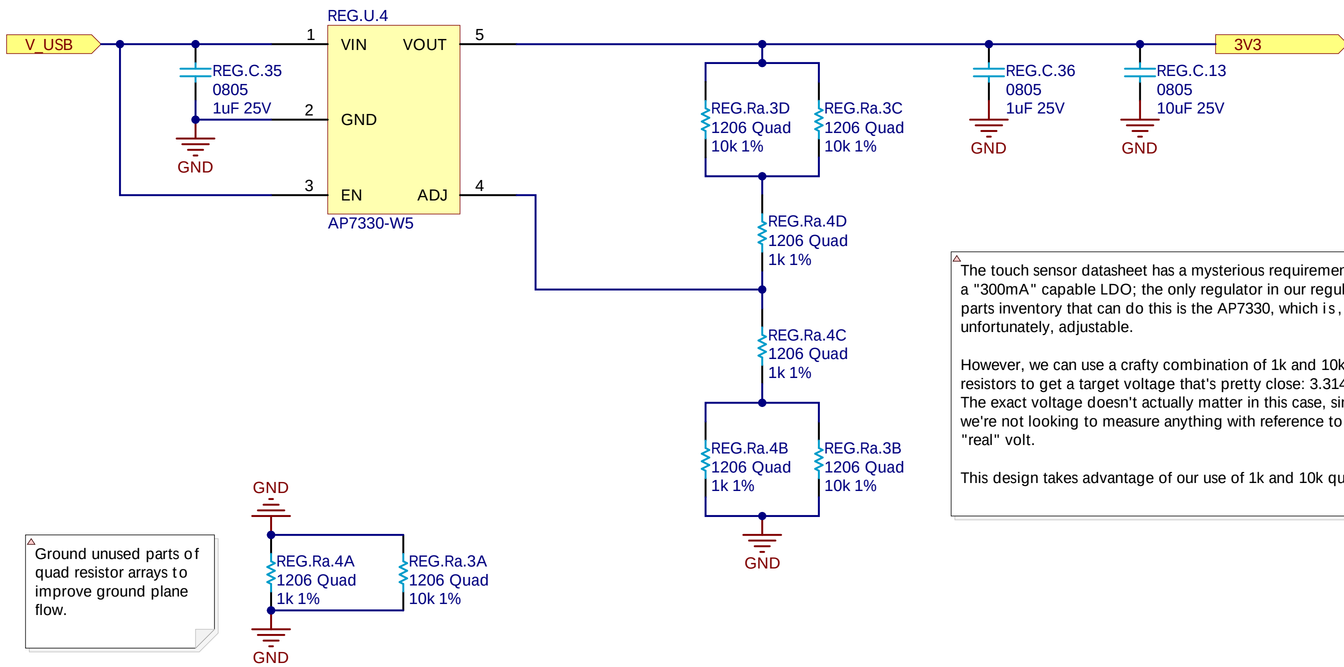

The touch sensor datasheet has a mysterious requirement for a "300mA" capable LDO; the only regulator in our regular parts inventory that can do this is the AP7330, which is, unfortunately, adjustable.

However, we can use a crafty combination of 1k and 10k resistors to get a target voltage that's pretty close: 3.314V. The exact voltage doesn't actually matter in this case, since we're not looking to measure anything with reference to a "real" volt.

This design takes advantage of our use of 1k and 10k quads.

Ground unused parts of quad resistor arrays to improve ground plane flow.

Title: Regulators.SchDoc

PCB: Trackpad.PrjPCB

Version: R2.002

Date: 10/23/2024 Time: 9:31:18 PM

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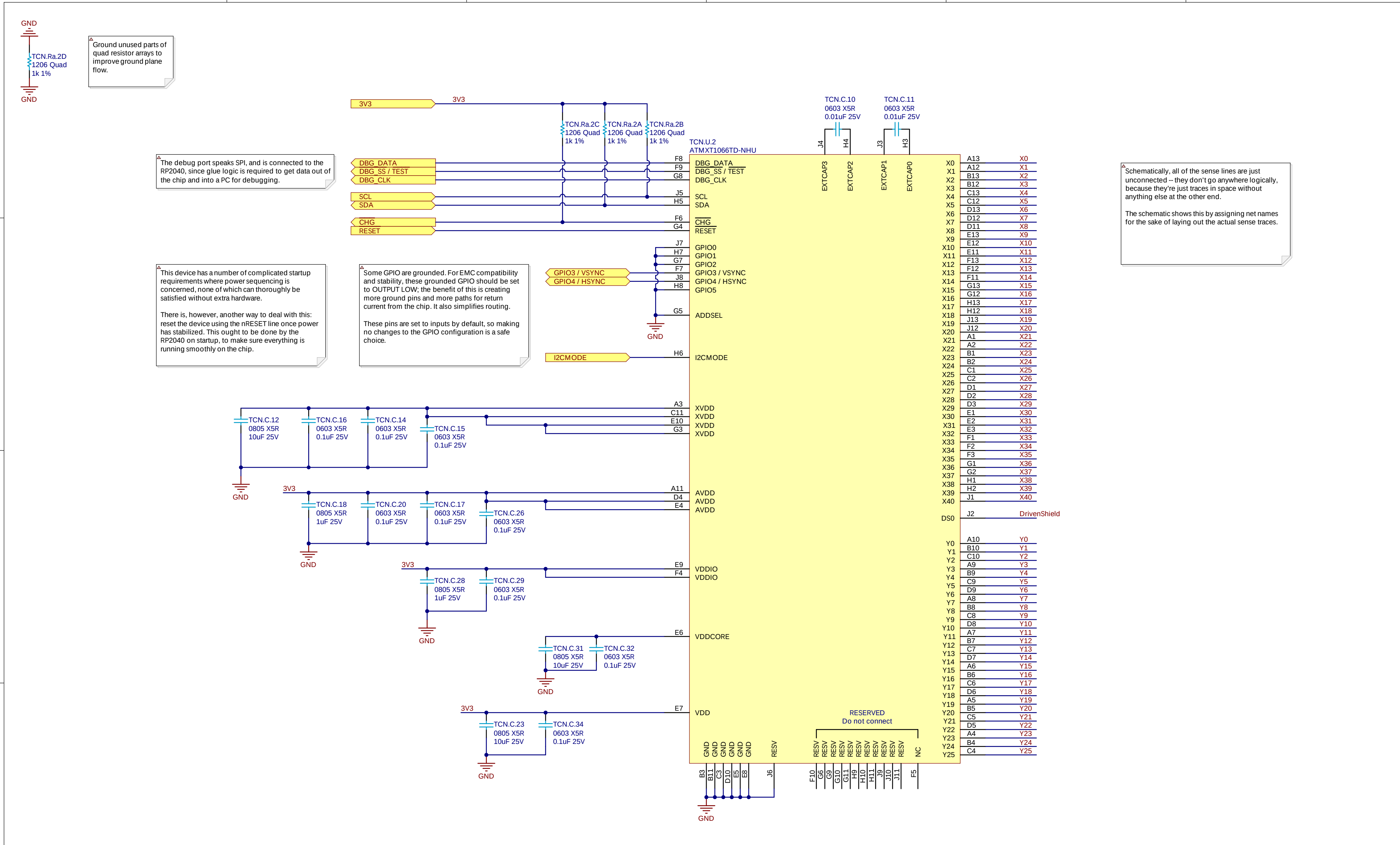

Title:	TouchController.SchDoc
PCB:	Trackpad.PrjPCB
Version:	R2.002
Date:	10/23/2024
Time:	9:31:18 PM
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